

## Chapter-3: Calculus of VVFs and VFs

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### 1. Vector-Valued functions:

- $\mathbf{r}(t) = \langle x(t), y(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j}$  is a vector-function in  $\mathbb{R}^2$ .

It traces a plane curve.

**Example:**  $\mathbf{r} = \langle a \cos t, a \sin t \rangle$  is a circle, plane curve, oriented counterclockwise.

- $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$  vector function in  $\mathbb{R}^3$ .

It traces a space curve.

**Example:**  $\mathbf{r} = \langle a \cos t, a \sin t, ct \rangle$  is a circular helix, a space curve.

- A line segment from point  $A$  to  $B$  is defined by the vector function

$$\mathbf{r}(t) = A + (B - A)t, \quad 0 \leq t \leq 1.$$

**Example:** the line segment from the point  $(1, -2, 0)$  to  $(2, -1, 3)$  :

$$\mathbf{r}(t) = (1, -2, 0) + (1, 1, 3)t, \quad 0 \leq t \leq 1.$$

- The part of the graph of  $y = f(x)$  from  $(a, f(a))$  to the point  $(b, f(b))$  is a plane curve defined by

$$\mathbf{r}(t) = \langle t, f(t) \rangle, \quad a \leq t \leq b.$$

**Example:** The part of the parabola  $y = x^2$  from  $(-1, 1)$  to  $(2, 4)$  is

$$\mathbf{r}(t) = \langle t, t^2 \rangle, \quad -1 \leq t \leq 2.$$

- The intersection of two surfaces is a curve.

**Example:** the intersection of the cylinder  $x^2 + y^2 = 4$  and the plane  $z = 2 + y$  is an elliptical space curve:

$$\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 2 + 2 \sin t \rangle \quad 0 \leq t \leq 2\pi,$$

with positive orientation (counterclockwise).

### 2. Derivatives of vector functions:

- If  $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ , then

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

- **Differentiation Rules:** Suppose  $\mathbf{u}$  and  $\mathbf{v}$  are differentiable vector functions,  $c$  is a scalar, and  $f$  is a real-valued function. Then

- (a)  $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
- (b)  $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
- (c)  $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
- (d)  $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
- (e)  $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
- (f)  $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

- A curve  $C$  defined by  $\mathbf{r} = \mathbf{r}(t)$ ,  $a \leq t \leq b$ , is **smooth** if  $\|\mathbf{r}'(t)\| \neq 0$  for all  $t \in (a, b)$
- $\mathbf{r}'(t)$  is **tangent** to the curve  $C$  at the point  $\mathbf{r}(t)$

**Example:** Show the the curve  $C : \mathbf{r}(t) = \langle t \sin 2t, t^2, t \cos 2t \rangle$  is **smooth**

**Solution:**  $\mathbf{r}'(t) = \langle \sin 2t + 2t \cos 2t, 2t, \cos 2t - 2t \sin 2t \rangle$  is continuous for all real number  $t$ , and

$$\begin{aligned} \|\mathbf{r}'\| &= \sqrt{\sin^2 2t + 4t \cos 2t \sin 2t + 4t^2 \cos^2 2t + 4t^2 + \cos^2 2t - 4t \cos 2t \sin 2t + 4t^2 \cos^2 2t} \\ &= \sqrt{1 + 8t^2} \neq 0, \forall t \in \mathbb{R}. \therefore, C \text{ is smooth} \end{aligned}$$

### 3. Integrals of vector functions:

- If  $\mathbf{R}'(t) = \mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ , then

$$\begin{aligned} \int \mathbf{r}(t) dt &= \int \langle x(t), y(t), z(t) \rangle dt = \left\langle \int x(t) dt + k_1, \int y(t) dt + k_2, \int z(t) dt + k_3 \right\rangle \\ &= \left\langle \int x(t) dt, \int y(t) dt, \int z(t) dt \right\rangle + \langle k_1, k_2, k_3 \rangle \\ &= \mathbf{R}(t) + \mathbf{k}, \quad k = \langle k_1, k_2, k_3 \rangle \text{ arbitrary constant vector} \end{aligned}$$

and

$$\begin{aligned} \int_a^b \mathbf{r}(t) dt &= \int_a^b \langle x(t), y(t), z(t) \rangle dt = \left\langle \int_a^b x(t) dt, \int_a^b y(t) dt, \int_a^b z(t) dt \right\rangle \\ &= \mathbf{R}(b) - \mathbf{R}(a) \end{aligned}$$

**Example:** Evaluate the integral  $\int_0^1 \left( \frac{4}{1+t^2} \mathbf{i} + \frac{2t}{1+t^2} \mathbf{j} \right) dt$

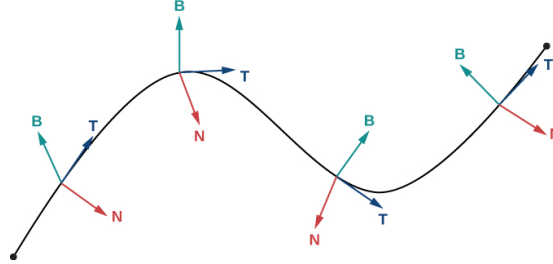
**Solution:**  $\int_0^1 \left( \frac{4}{1+t^2} \mathbf{i} + \frac{2t}{1+t^2} \mathbf{j} \right) dt = \int_0^1 \frac{4}{1+t^2} dt \mathbf{i} + \int_0^1 \frac{2t}{1+t^2} dt \mathbf{j}$

$$= 4 \tan^{-1} t \Big|_0^1 \mathbf{i} + \ln(1+t^2) \Big|_0^1 \mathbf{j} = \pi \mathbf{i} + \ln 2 \mathbf{j}$$

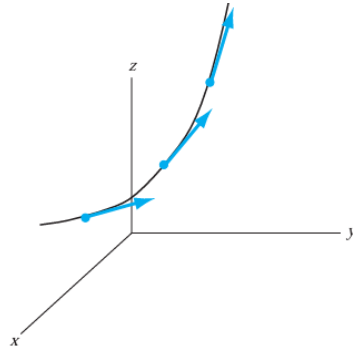
4. Let  $C : \mathbf{r} = \mathbf{r}(t)$ ,  $a \leq t \leq b$  be a **smooth** curve. Then

- **Arc-length** of  $C$  is  $L = \int_a^b \|\mathbf{r}'(t)\| dt$
- **Arc-length function** of  $C$  is  $s(t) = \int_a^t \|\mathbf{r}'(u)\| du$ ,  $a < t < b$
- **Differential** of arc-length function:  $ds = \|\mathbf{r}'\| dt$ .

- **Unit tangent vector** at  $t$  is  $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ .
  - $\mathbf{T}$  is tangent to  $C$ .
  - $\mathbf{T}$  is a unit vector.
  - $\mathbf{T}$  points in the direction of the increasing parameter  $t$ .



- **Unit normal vector** at  $t$  is  $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$ ,  $\|\mathbf{T}'(t)\| \neq 0$ .
  - $\mathbf{N}$  is undefined if  $\|\mathbf{T}'(t)\| = 0$
  - $\mathbf{N}$  points to the concave side of  $C$
  - $\mathbf{N} \perp \mathbf{T} \Rightarrow \mathbf{T} \cdot \mathbf{N} = 0$ .
  - $\mathbf{N}$  is a unit vector.
- **Curvature** of  $C$  at  $t$  is  $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$



The more a curve bends at a point, the faster the unit tangent vector is changing direction there.

- $\kappa$  is a scalar and  $\kappa \geq 0$ .
- $\kappa = 0 \Rightarrow \mathbf{T}$  does not change. That is  $C$  is a straight line.
- $\kappa$  is very large means  $C$  is very bending.
- $\kappa$  close to zero means  $C$  is nearly a straight line.
- curvature of a circle of radius  $a$  is  $\frac{1}{a}$ .
- **Radius of curvature** at  $t$  is  $r(t) = \frac{1}{\kappa(t)}$ .

**Example:** For the curve  $C: \mathbf{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ ,  $0 \leq t \leq 2\pi$ , find

*arclength(L), unit tangent vector(T), unit normal vector(N)  
curvature( $\kappa$ ) and radius of curvature(r).*

**Solution:**  $\mathbf{r}'(t) = \langle e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t, e^t \rangle$  and  $\|\mathbf{r}'(t)\| = \sqrt{3}e^t$ .

- $L = \int_0^{2\pi} \|\mathbf{r}'(t)\| dt = \int_0^{2\pi} \sqrt{3}e^t dt = \sqrt{3} [e^{2\pi} - 1]$
- $s(t) = \int_0^t \|\mathbf{r}'(u)\| du = \int_0^t \sqrt{3}e^u du = \sqrt{3} (e^t - 1)$  and
- $ds = \|\mathbf{r}'(t)\| dt = \sqrt{3}e^t dt$
- $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \left\langle \frac{\cos t - \sin t}{\sqrt{3}}, \frac{\sin t + \cos t}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$ .
- $\mathbf{T}'(t) = \left\langle \frac{-\cos t - \sin t}{\sqrt{3}}, \frac{\cos t - \sin t}{\sqrt{3}}, 0 \right\rangle$  and  $\|\mathbf{T}'(t)\| = \frac{\sqrt{6}}{3}$
- $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \left\langle \frac{-\cos t - \sin t}{\sqrt{2}}, \frac{\cos t - \sin t}{\sqrt{2}}, 0 \right\rangle$ .
- $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\frac{\sqrt{6}}{3}}{\sqrt{3}e^t} = \frac{\sqrt{2}}{3} e^{-t}$
- $r(t) = \frac{1}{\kappa(t)} = \frac{3e^t}{\sqrt{2}}$

##### 5. Vector-fields: Examples of vector field

- **Velocity vector fields.**
  - moving fluid such as air or water.
- **Force vector-field:**
  - gravitational force  $\rightarrow \mathbf{F}(\mathbf{r}) = -\frac{mMG}{\|\mathbf{r}\|^3} \mathbf{r}$ ,
  - electric force  $\rightarrow \mathbf{F}(\mathbf{r}) = \frac{\epsilon qQ}{\|\mathbf{r}\|^3} \mathbf{r}$ ,
- **Gradient vector-field**  $\rightarrow \nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$
- A vector field  $\mathbf{F}$  on  $\mathbb{R}^3$  is defined by
 
$$\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle$$

$$= M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$
 where  $M, N, P$  are scalar functions also called **scalar fields** in  $\mathbb{R}^3$   
**For example,**  $\mathbf{F}(x, y, z) = \langle x + yz^2, 3y - xz, xye^z \rangle$ , is a vector field in  $\mathbb{R}^3$  with component functions  $M = xyz^2, N = 3y - xz, P = xye^z$
- A vector field  $\mathbf{F}$  on  $\mathbb{R}^2$  is defined by
 
$$\mathbf{F}(x, y) = \langle M(x, y), N(x, y) \rangle = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$$
- The best way to picture a vector field is to draw the arrow representing the vector  $\mathbf{F}(x, y)$  starting at the point  $(x, y)$ .

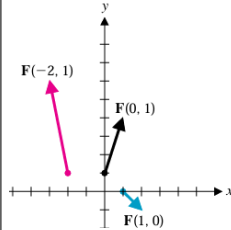
**Example:** For the vector field  $\mathbf{F}(x, y) = \langle x + y, 3y - x \rangle$ , evaluate and plot  $\mathbf{F}(1, 0)$ ,  $\mathbf{F}(0, 1)$ ,  $\mathbf{F}(-2, 1)$

**Solution:**  $\mathbf{F}(1, 0) = \langle 1, -1 \rangle$ . We can plot the vector  $\langle 1, -1 \rangle$  with its initial point at  $(1, 0)$ , and terminal point at  $(2, -1)$ .

$\mathbf{F}(0, 1) = \langle 1, 3 \rangle$ . Plot the vector  $\langle 1, 3 \rangle$ , taking  $(0, 1)$  as its

initial point, and its terminal point is at  $(1, 4)$ .

- $\mathbf{F}(-2, 1) = \langle -1, 5 \rangle$ . Plot  $\langle -1, 5 \rangle$  by placing its initial point at  $(-2, 1)$  and its terminal point is at  $(-1, 5)$ .



## 6. Conservative Vector Field:

- A vector field  $\mathbf{F}$  is **conservative** if  $\mathbf{f} = \nabla f$  for some scalar field  $f$ .

We call  $f$  a potential function, or potential, for  $\mathbf{F}$ .

Of course, if  $f$  is a potential, so is  $f + c$ , for any constant  $c$ .

**For example,**  $\mathbf{F} = \langle y^2 z^2 - 1, 2xyz^2, 4z^3 \rangle$  is a conservative vector field since there is a scalar function  $f(x, y, z) = xy^2 z^2 - x + z^4$  such that  $\nabla f = \langle y^2 z^2 - 1, 2xyz^2, 4z^3 \rangle = \mathbf{F}$ .

## 7. Divergence and Curl:

Let  $\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$  be a vector field.

- **Divergence** of  $\mathbf{F}$  is the scalar field:  $\text{div } \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}$ .
- **Curl** of  $\mathbf{F}$  is the vector field  $\text{curl } \mathbf{F} = \left( \frac{\partial P}{\partial y} - \frac{\partial N}{\partial z} \right) \mathbf{i} + \left( \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x} \right) \mathbf{j} + \left( \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \mathbf{k}$ .
- the **del operator**:  $\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$  is **differential operator**.
  - The **product** of the vector  $\nabla$  and the scalar function  $f$  is the gradient of  $f$ :  $\nabla f = \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$ .
  - The **dot product** of  $\nabla$  and  $\mathbf{F}$  is the divergence of  $\mathbf{F}$ :
 
$$\begin{aligned} \nabla \cdot \mathbf{F} &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (M\mathbf{i} + N\mathbf{j} + P\mathbf{k}) \\ &= \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = \text{div } \mathbf{F} \end{aligned}$$
  - The **cross product** of  $\nabla$  with  $\mathbf{F}$  is the curl of  $\mathbf{F}$ :
 
$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix} = \text{curl } \mathbf{F}.$$
- There are two relationships between gradient, divergence and curl that are fundamental to vector analysis:
  - (a) the **curl** of a **gradient** is the **zero vector**,  $\nabla \times \nabla f = \mathbf{0}$  ( $\text{curl grad} = \mathbf{0}$ .)
  - (b) the **divergence** of a **curl** is (the number) **zero**,  $\nabla \cdot (\nabla \times \mathbf{F}) = 0$  ( $\text{div curl} = 0$ .)

**Example:** Find the divergence and curl of  $\mathbf{F}(x, y, z) = xz\mathbf{i} + xyz\mathbf{j} - y^2\mathbf{k}$ .

**Solution:**  $\text{div } \mathbf{F} = M_x + N_y + P_z = z + xz$

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} = \langle -2y - xy, x, yz \rangle$$

### 8. Test for Conservative Vector-Field:

- Let  $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$  be a vector field whose component functions have continuous partial derivatives.  $\mathbf{F}$  is a conservative if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$$

- To find a **potential function**  $f$  for a conservative vector field  $\mathbf{F}$ , solve the following system of equations for  $f$ :

$$\begin{cases} f_x = M(x, y, z) & (1) \\ f_y = N(x, y, z) & (2) \\ f_z = P(x, y, z) & (3) \end{cases}$$

**Example:** Show that  $\mathbf{F} = \langle e^x \cos y + yz, xz - e^x \sin y, xy - z \rangle$  is conservative and find a potential function for it

**Solution:** Let  $M = e^x \cos y + yz$ ,  $N = xz - e^x \sin y$ ,  $P = xy - z$ .

$$P_y = x = N_z, \quad M_z = y = P_x, \quad N_x = -e^x \sin y = M_y. \Rightarrow \mathbf{F} \text{ is conservative.}$$

- $\mathbf{F}$  is **conservative**, there is a **potential function**  $f$  with  $\mathbf{F} = \nabla f$ .

$$\begin{cases} f_x = e^x \cos y + yz & (1) \\ f_y = xz - e^x \sin y & (2) \\ f_z = xy - z & (3) \end{cases}$$

- Integrating (1) with respect to  $x$ , (2) with respect to  $y$  and (3) with respect to  $z$ , respectively, we get

$$\begin{cases} f = e^x \cos y + xyz + g(x, y) & (4) \\ f = xyz + e^x \cos y + h(x, z) & (5) \\ f = xyz - \frac{z^2}{2} + k(x, y) & (6) \end{cases}$$

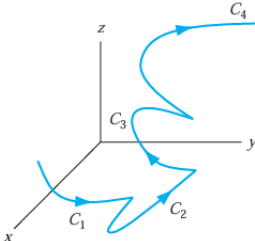
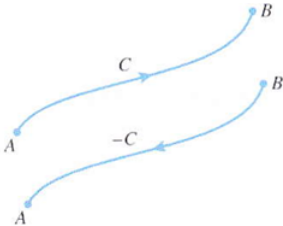
- From (4), (5) and (6), we chose  $f(x, y, z) = e^x \cos y + xyz - \frac{z^2}{2}$ .

### 9. Line Integral of a scalar function: $\int_C f ds$ and $\int_C \mathbf{F} \cdot d\mathbf{r}$

- $\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$ ,  $C: \mathbf{r} = \mathbf{r}(t)$ ,  $a \leq t \leq b$  is a smooth.

$$- \int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left[\frac{dx}{dt}\right]^2 + \left[\frac{dy}{dt}\right]^2} dt$$

$$- \int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left[\frac{dx}{dt}\right]^2 + \left[\frac{dy}{dt}\right]^2 + \left[\frac{dz}{dt}\right]^2} dt$$

- **Steps to evaluate  $\int_C f ds$  :**
  - Represent  $C$  by a vector function  $\mathbf{r}(t)$  for  $a \leq t \leq b$  if not.
  - Find  $f(\mathbf{r}(t))$  (value of  $f$  along  $C$ ).
  - Find  $\mathbf{r}'(t)$  and then  $ds = \|\mathbf{r}'(t)\| dt$
  - Evaluate the line integral:  $\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$
- **Properties of  $\int_C f ds$  :**
  - (i) If  $C = C_1 \cup \dots \cup C_n$ , where  $C_1, \dots, C_n$  are smooth curves and the initial point of  $C_{i+1}$  is the terminal point of  $C_i$ . Then
$$\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds + \dots + \int_{C_n} f ds$$


  - (ii)  $\int_C f ds = \int_{-C} f ds$ ,  $-C$  denotes the curve consisting of the same points as  $C$  but with the opposite orientation.
  - (iii) For any constant  $k$ ,  $\int_C k f ds = k \int_C f ds$ .
  - (iv) For  $f = 1$ ,  $\int_C f ds = \int_C ds$  gives the arc length of the curve  $C$ .
- **For example,**
  - a) If the arclength of  $C$  is 7 and  $f = 3$  then  $\int_C f ds = \dots\dots$
  - b) If the arclength of  $C$  is  $2\pi$  and  $f = -\sqrt{6}$  then  $\int_C f ds = \dots\dots$

**Example:** Evaluate  $\int_C 2y\sqrt{1+x^2} ds$ ,  $C$  consists of the parabola  $y = \frac{x^2}{2}$  from  $(0, 0)$  to  $(1, \frac{1}{2})$  and the line segment from  $(1, \frac{1}{2})$  to  $(1, \frac{3}{2})$ .

**Solution:**  $C = C_1 \cup C_2$ ,  $C_1 : \mathbf{r}_1(t) = \left\langle t, \frac{t^2}{2} \right\rangle$ ,  $0 \leq t \leq 1$  and  
 $C_2 : \mathbf{r}_2(t) = \left\langle 1, \frac{1}{2} + t \right\rangle$ ,  $0 \leq t \leq 1$ .

- $f(\mathbf{r}_1(t)) = t^2 \sqrt{1+t^2}$  and  $f(\mathbf{r}_2(t)) = \sqrt{2}(2t+1)$ .
- $\mathbf{r}'_1(t) = \langle 1, t \rangle$  and  $\mathbf{r}'_2(t) = \langle 0, 1 \rangle$   
 $f(\mathbf{r}_1(t)) \|\mathbf{r}'_1(t)\| = t^2 + t^4$  and  $f(\mathbf{r}_2(t)) \|\mathbf{r}'_2(t)\| = \sqrt{2}(2t+1)$ .
- $\int_{C_1} f ds = \int_0^1 (t^2 + t^4) dt = \left( \frac{t^3}{3} + \frac{t^5}{5} \right) \Big|_0^1 = \frac{8}{15}$  and  
 $\int_{C_2} f ds = \int_0^1 \sqrt{2}(2t+1) dt = \sqrt{2} (t^2 + t) \Big|_0^1 = 2\sqrt{2}$
- $\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds = \frac{8+30\sqrt{2}}{15}$

- Suppose that  $\rho(x, y, z)$  represents the linear density at a point  $(x, y, z)$  of a thin wire shaped like a curve  $C$ . Then the

(i) **mass**  $m$  is  $m = \int_C \rho(x, y, z) ds$

(ii) the **center of mass** is  $(\bar{x}, \bar{y}, \bar{z})$ , where

$$\bar{x} = \frac{1}{m} \int_C x \rho ds, \quad \bar{y} = \frac{1}{m} \int_C y \rho ds, \quad \bar{z} = \frac{1}{m} \int_C z \rho ds$$

**Example:**

Find the mass and center of mass of a wire in the shape of the helix  $x = t$ ,  $y = \cos t$ ,  $z = \sin t$ ,  $0 \leq t \leq 2\pi$  if the density at any point is equal to the square of the distance from the origin.

**Solution:** Let  $P(x, y, z)$  be a point on the wire, then the density is

$$\rho(x, y, z) = x^2 + y^2 + z^2 = t^2 + \cos^2 t + \sin^2 t = t^2 + 1.$$

$$ds = \sqrt{(1)^2 + (-\sin t)^2 + (\cos t)^2} dt = \sqrt{2} dt$$

$$\bullet m = \int_C \rho ds = \int_0^{2\pi} (t^2 + 1) \sqrt{2} dt = \sqrt{2} \left( \frac{t^3}{3} + t \right) \Big|_0^{2\pi} = \frac{\sqrt{8} (4\pi^2 + 3) \pi}{3}$$

• Let  $(\bar{x}, \bar{y}, \bar{z})$  the center of mass of the wire, where

$$\begin{aligned} \bar{x} &= \frac{1}{m} \int_C x \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^3 + t) \sqrt{2} dt \\ &= \frac{3}{2\pi(4\pi^2 + 3)} \left( \frac{t^4}{4} + \frac{t^2}{2} \right) \Big|_0^{2\pi} = \frac{6\pi^3 + 3}{4\pi^2 + 3} \end{aligned}$$

$$\bar{y} = \frac{1}{m} \int_C y \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^2 + 1) \cos t \sqrt{2} dt$$

$$= \frac{3}{2\pi(4\pi^2 + 3)} (t^2 \sin t + 2t \cos t - \sin t) \Big|_0^{2\pi} = \frac{6}{4\pi^2 + 3}$$

$$\bar{z} = \frac{1}{m} \int_C z \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^2 + 1) \sin t \sqrt{2} dt$$

$$= \frac{3}{2\pi(4\pi^2 + 3)} (-t^2 \cos t + 2t \sin t + \cos t) \Big|_0^{2\pi} = \frac{-6\pi}{4\pi^2 + 3}$$

10. Line Integral of a Vector field:  $\int_C \mathbf{F} \cdot d\mathbf{r}$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_C M dx + N dy + P dz$$

$$= \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$$

- If  $\mathbf{F}$  is a force-field, the work done by  $\mathbf{F}$  in moving a particle along the curve  $C$  can be written simply as

$$W = \int_C \mathbf{F}(x, y, z) \cdot d\mathbf{r}$$

- If  $\mathbf{F}$  a velocity field of a fluid flowing through a region in space, then the flow along the curve  $C$  is

$$flow = \int_C \mathbf{F} \cdot d\mathbf{r}$$

The integral is called a **flow integral**.

- If  $C$  is a closed loop, the flow is called the **circulation** around  $C$ .

- **Steps to evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$  :**

- Represent  $C$  as a vector function  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ , if not.
- Find  $\mathbf{F}(\mathbf{r}(t))$ ,  $\mathbf{r}'(t)$  and  $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$ .
- Evaluate  $\int_a^b [\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)] dt$

- **Properties of  $\int_C \mathbf{F} \cdot d\mathbf{r}$  :**

- Reversing the orientation of  $C$  reverses the sign of the integral; that is,

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}$$

- If  $C$  is made by joining a finite number  $C_1, \dots, C_n$  of curves end to end, then  $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \dots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}$

#### Example:

Compute the work done by the force field  $\mathbf{F} = \langle 4y, 2xz, 3y \rangle$  acting on an object as it moves along the helix defined by  $x = 2 \cos t$ ,  $y = 2 \sin t$ ,  $z = 3t$ , from the point  $(2, 0, 0)$  to  $(-2, 0, 3\pi)$ .

**Solution:**  $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C 4y dx + 2xz dy + 3y dz.$

- The point  $(2, 0, 0)$  corresponds to  $t = 0$  and  $(-2, 0, 3\pi)$  corresponds to  $t = \pi$ .
- $dx = -2 \sin t dt$ ,  $dy = 2 \cos t dt$  and  $dz = 3 dt$ .

$$\begin{aligned} W &= \int_0^\pi [-16 \sin^2 t + 24t \cos^2 t + 18 \sin t] dt \\ &= \int_0^\pi [-8 + 8 \cos 2t + 12t + 12t \cos 2t + 18 \sin t] dt \\ &= 36 - 8\pi + 6\pi^2 \end{aligned}$$

#### 11. The Fundamental Theorem of Line Integrals: (FTLI)

Let  $C : \mathbf{r} = \mathbf{r}(t)$ ,  $a \leq t \leq b$  be a smooth curve. Let  $f$  be a differentiable function of two or three variables whose gradient vector  $\nabla f = \mathbf{F}$  is continuous on  $C$ . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

**Example: Evaluate**

a)  $\int_{(0,0,0)}^{(3,3,1)} 2x dx - y^2 dy - \frac{4}{1+z^2} dz$

b)  $\int_C e^y dx + x e^y dy + (1+z) e^z dz, \quad C: \mathbf{r}(t) = \langle t, t^2, t^3 \rangle, \quad 0 \leq t \leq 1.$

**Solution:** (a)  $\mathbf{F} = \langle 2x, -y^2, -\frac{4}{1+z^2} \rangle$  and  $f = x^2 - \frac{y^3}{3} - 4 \tan^{-1} z$  with  $\mathbf{F} = \nabla f$ .

•  $\int_{(0,0,0)}^{(3,3,1)} 2x dx - y^2 dy - \frac{4}{1+z^2} dz = \left( x^2 - \frac{y^3}{3} - 4 \tan^{-1} z \right) \Big|_{(0,0,0)}^{(3,3,1)} = -\pi$

**Solution:** (b)  $\mathbf{F} = \langle e^y, x e^y, (1+z) e^z \rangle$  and  $f = x e^y + z e^z$  with  $\mathbf{F} = \nabla f$

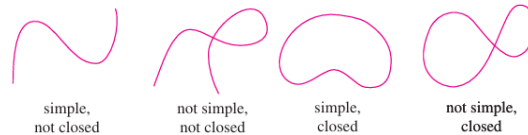
•  $\mathbf{r}(0) = (0, 0, 0)$  and  $\mathbf{r}(1) = (1, 1, 1)$ .

• By FTLI,  $\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) = 2e$

$\int_C \mathbf{F} \cdot d\mathbf{r}$  is independent of path in  $D$  (region) if and only if  $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$  for every closed path in  $D$ .

## 12. Closed curve, simple curve, simple closed curve:

- A curve is called closed if its terminal point coincides with its initial point, that is,  $\mathbf{r}(a) = \mathbf{r}(b)$ .
- A **simple curve** is a curve that doesn't intersect itself anywhere between its endpoints.
- For a **simple closed curve**,  $\mathbf{r}(a) = \mathbf{r}(b)$  but  $\mathbf{r}(t_1) \neq \mathbf{r}(t_2)$  when  $a < t_1 < t_2 < b$ .



## 13. Green's Theorem: $\oint_C M dx + N dy = \iint_D (N_x - M_y) dA$

Let  $C$  be a positively oriented, piecewise-smooth, simple closed curve in the plane and let  $D$  be the region bounded by  $C$ . If  $M$ ,  $N$ ,  $\frac{\partial M}{\partial y}$  and  $\frac{\partial N}{\partial x}$  be continuous on an open region that contains  $D$ , then

$$\int_C M dx + N dy = \iint_D (N_x - M_y) dA$$

**Example: Use Green's Theorem**

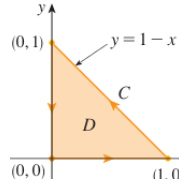
- (a) Find the work done by  $\mathbf{F} = (y - x^2 e^x)\mathbf{i} + (\cos(2y^2) - x)\mathbf{j}$  in moving a particle counter clockwise about the rectangular path  $C$  having vertices  $(0, 1)$ ,  $(1, 1)$ ,  $(1, 3)$  and  $(0, 3)$ .

**Solution:**  $M = y - x^2 e^x, \quad N = \cos(2y^2) - x \Rightarrow N_x - M_y = -2$

$$Work = \oint_C \mathbf{f} \cdot d\mathbf{r} = \iint_D (N_x - M_y) dA = \iint_D -2 dA = -2 \text{ Area}(D) = -4.$$

- (b) Evaluate  $\int_C x^4 dx + xy dy$ , where  $C$  is the triangular curve consisting of the line segments from  $(0, 0)$  to  $(1, 0)$ , from  $(1, 0)$  to  $(0, 1)$ , and from  $(1, 0)$  to  $(0, 0)$ .

**Solution:**  $M = x^4$ ,  $N = xy \Rightarrow N_x - M_y = y$



$$D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (N_x - M_y) dA = \int_0^1 \int_0^{1-x} y dx dy = \frac{1}{2} \int_0^1 (1-x)^2 dx = \frac{1}{6}$$

- (c) Evaluate  $\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{1+y^4}) dy$ , where  $C$  is the circle  $x^2 + y^2 = 9$ .

**Solution:**  $M = 3y - e^{\sin x}$ ,  $N = 7x + \sqrt{1+y^4} \Rightarrow N_x - M_y = 7 - 3 = 4$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (N_x - M_y) dA = \iint_D 4 dA = 4 \text{ Area}(D) = 36\pi.$$

- (d) Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$ ,  $\mathbf{F} = \langle \sqrt{x} + y^3, x^2 + \sqrt{y} \rangle$  and  $C$  consists of the arc of the curve  $y = \sin x$  from  $(0, 0)$  to  $(\pi, 0)$  and the line segment from  $(\pi, 0)$  to  $(0, 0)$ .

**Solution:**  $M = \sqrt{x} + y^3$ ,  $x^2 + \sqrt{y} \Rightarrow N_x - M_y = 2x - 3y^2$

$$D = \{(x, y) \mid 0 \leq x \leq \pi, 0 \leq y \leq \sin x\}$$

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= - \iint_D (N_x - M_y) dA = \int_0^\pi \int_0^{\sin x} (-2x + 3y^2) dy dx \\ &= \int_0^\pi [-2xy + y^3]_0^{\sin x} dx = \int_0^\pi [-2x \sin x + \sin^3 x] dx \\ &= \left[ 2x \cos x + 2 \sin x - \frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x \right]_0^\pi = \frac{4}{3} - 2\pi \end{aligned}$$

- (e) A particle starts at the point  $(-2, 0)$ , moves along the  $x$ -axis to  $(2, 0)$ , and then along the semicircle  $y = \sqrt{4 - x^2}$  to the starting point. Find the work done on this particle by the force field  $\mathbf{F} = \langle x, y^3 + 3xy^2 \rangle$ .

**Solution:**  $M = x$ ,  $N = y^3 + 3xy^2 \Rightarrow N_x - M_y = 3y^2 = 3r^2 \sin^2 \theta$ .

$$D = [0, 2] \times [0, \pi] \text{ and } dA = r dr d\theta$$

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (N_x - M_y) = \int_0^{2\pi} \int_0^2 3r^3 \sin^2 \theta dr d\theta \\ &= 8 \int_0^{2\pi} \sin^2 \theta d\theta = 8 \left[ -\frac{1}{2} \sin \theta \cos \theta + \frac{\theta}{2} \right]_0^{2\pi} = 8\pi. \end{aligned}$$

**Exercises:**

- 1) Find curvature, unit tangent and unit normal vectors of the
  - (a)  $\mathbf{r}(t) = 3e^{-t}(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ ,      (b)  $\mathbf{r}(t) = e^t \sin t \mathbf{i} - \mathbf{j} + e^t \cos t \mathbf{k}$
- 2) Compute  $\nabla \cdot \mathbf{F}$  and  $\nabla \times \mathbf{F}$  and verify explicitly that  $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ .
  - (a)  $\mathbf{F} = \sinh(xyz)\mathbf{z}$       (b)  $\mathbf{F} = 2xy\mathbf{i} + xe^y\mathbf{j} + 2z\mathbf{k}$
- 3) Compute  $\nabla f$  and verify explicitly that  $\nabla \times (\nabla f) = \mathbf{0}$ 
  - (a)  $f = e^{x+y+z}$       (b)  $f = \sin(xz)$
- 4) Find the mass and center of mass of a thin, straight wire extending from the origin to  $(3, 3, 3)$  if  $\sigma = x + y + z$  grams per centimeter.
- 5) Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$ , where
  - i.  $\mathbf{F}(x, y) = e^{4y} (2x\mathbf{i} + 4x^2\mathbf{j})$ , and  $C$  is the portion of the parabola  $y = x^2 - 4$  from  $(0, 0)$  to  $(2, 0)$
  - ii.  $\mathbf{F}(x, y) = e^{x^2+y^2+z^2} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$  and  $C$  is the line segment from  $(0, 0, 0)$  to  $(1, 1, 0)$ .
  - iii.  $\mathbf{F} = \langle x^2 - y, \cos 2y - e^{3y} + 4x \rangle$ ,  $C$  is a square with sides of length 5.
  - iv.  $\mathbf{F} = e^x \cos y \mathbf{i} - e^x \sin y \mathbf{j}$  and  $C$  is any simple closed path.
- 6) A particle moves once counterclockwise around the circle of radius 6 about the origin, under the influence of the force  $\mathbf{F} = (e^x - y + x \cosh(x))\mathbf{i} + (y^{\frac{3}{2}} + x)\mathbf{j}$ . Calculate the work done.

14. **Representation of Surfaces in Space:**

- Explicit form  $S: z = g(x, y)$ .
  - (i)  $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle$  **upward normal vector.**
  - (ii)  $dS = \|\mathbf{N}\| dA = \sqrt{1 + g_x^2 + g_y^2} dx dy$
  - (iii) *surface area of  $S$*   $= \iint_S dS = \iint_D \|\mathbf{N}\| dA = \iint_D \sqrt{1 + g_x^2 + g_y^2} dx dy$

**For example  $S: z = x^2 + y^2, 0 \leq z \leq 2$ .**

  - $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle = \langle -2x, -2y, 1 \rangle$
  - $dS = \|\mathbf{N}\| dA = 2\sqrt{x^2 + y^2} dA$
  - $A(S) = \iint_S dS = \iint_S 2\sqrt{x^2 + y^2} dA = 2 \int_0^{2\pi} \int_0^2 r r dr d\theta = \frac{32\pi}{3}$
- Implicit form  $S: g(x, y, z) = 0$ 

**For example  $S: x^2 + y^2 + z^2 = 4$ .**
- Parametric or vector form:
 
$$S: x = x(u, v), \quad y = y(u, v), \quad z = z(u, v),$$

or  $S: \mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \quad a \leq u \leq b, \quad c \leq v \leq d.$

  - $\mathbf{r}(u, v)$  is a vector from the origin to the point  $(x(u, v), y(u, v), z(u, v))$  on the surface  $S$ .

**Example-1:** A sphere  $S: x^2 + y^2 + z^2 = a^2$ .

- **Spherical coordinates:**  $x = a \sin \phi \cos \theta$ ,  $y = a \sin \phi \sin \theta$ ,  $z = a \cos \phi$ .
- **Parameters:**  $\phi = u$  and  $\theta = v$ .
- **Vector form:**  
 $S: \mathbf{r}(u, v) = \langle a \sin u \cos v, a \sin u \sin v, a \cos u \rangle$ ,  $(u, v) \in [0, \pi] \times [0, 2\pi]$
- **Normal vector:**  $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = a^2 \sin u \underbrace{\langle \sin u \cos v, \sin u \sin v, \cos u \rangle}_{\text{unit sphere}}$
- **Outward (positive) unit normal vector**  
 $\mathbf{n} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} = \langle \sin u \cos v, \sin u \sin v, \cos u \rangle$
- **Inward (negative) unit normal**  $\mathbf{n}' = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} = -\mathbf{n}$ .
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = a^2 \sin u du dv$
- *Surface area of  $S$*   $= \iint_S dS = \int_0^{2\pi} \int_0^\pi a^2 \sin u du dv = 4\pi a^2$

**Example-2:** A cylinder  $S: x^2 + y^2 = a^2$ ,  $h \leq z \leq H$

- We use **cylindrical coordinates:**  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = z$ .
- **Parameters:**  $\theta = u$  and  $z = v$ .
- **Vector form:**  $S: \mathbf{r}(u, v) = \langle a \cos u, a \sin u, v \rangle$ ,  $(u, v) \in [0, 2\pi] \times [h, H]$ .
- **Normal vector:**  $\mathbf{N} = \langle a \cos u, a \sin u, 0 \rangle$  **positive normal vector.**
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = a du dv$ .
- *Surface area of  $S$*   $= \iint_S dS = \int_0^{2\pi} \int_h^H a dv du = 2\pi a(H - h)$ .

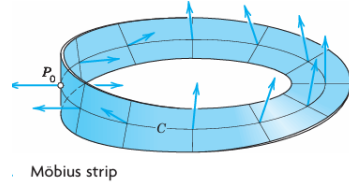
**Example-3:** A cone  $S: z = \sqrt{x^2 + y^2}$   $h \leq z \leq H$

- We use **cylindrical coordinates:**  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = z$ .
- **Parameters:**  $\theta = u$  and  $z = r = v$ .
- **Vector form:**  $S: \mathbf{r}(u, v) = \langle v \cos u, v \sin u, v \rangle$ ,  $(u, v) \in [0, 2\pi] \times [h, H]$ .
- **Normal vector:**  $\mathbf{N} = \langle v \cos u, v \sin u, -v \rangle$  **positive normal vector.**
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = \sqrt{2} v du dv$ .
- *Surface area of  $S$*   $= \iint_S dS = \int_0^{2\pi} \int_h^H \sqrt{2} v dv du = 2\pi\sqrt{2}\pi(H^2 - h^2)$ .

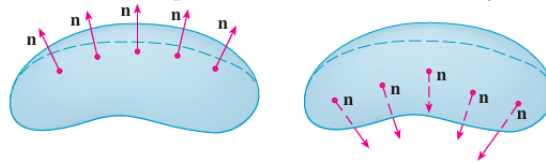
15. **Smooth and oriented surfaces:**

- If  $\mathbf{r}_u \times \mathbf{r}_v$  is not  $\mathbf{0}$ , then the surface  $S$  is called **smooth** (it has no “corners”).
- A surface is **simple** if it does not fold over and intersect itself.
- **Nonorientable** surface has one side.

For example, a Möbius strip has only one side - nonorientable.



- A surface  $S$  is **orientable** (or two-sided) if it is possible to define a unit normal vector  $\mathbf{n}$  at each point  $(x, y, z)$  not on the boundary of the surface and if  $\mathbf{n}$  is a continuous function of  $(x, y, z)$ .
- There are two possible orientations for any orientable surface.



- For a closed surface  $S$ ,
  - **positive orientation**=normal vectors pointing **outward** from  $S$
  - **negative orientation**=normal vectors pointing **inward** from  $S$

#### 16. Surface Integral of a Scalar Field: $\iint_S f(x, y, z) dS$

- Give a **smooth** surface

$S : \mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad D : a \leq u \leq b, \quad c \leq v \leq d$   
and a continuous function  $f(x, y, z)$  defined on  $S$ .

The **surface integral** of  $f$  on  $S$ , denoted by  $\iint_S f dS$ , is defined as

$$\begin{aligned} \int_S f dS &= \iint_D f(\mathbf{r}(u, v)) \|\mathbf{N}\| du dv \\ &= \int_a^b \int_c^d f(x(u, v), y(u, v), z(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dv du \end{aligned}$$

- For  $S : z = g(x, y), \quad (x, y) \in D$ ,

$$\iint_S f dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} dA,$$

$D$  = projection of  $S$  on the  $xy$ -plane.

#### Example-1:

Evaluate  $\iint_S 3z dS$ ,  $S$  is the portion of the plane  $x + y + z = 2$  inside the cylinder  $x^2 + y^2 = 1$

**Solution:**  $\mathbf{N} = \nabla(x + y + z) = \langle 1, 1, 1 \rangle$  and  $\|\mathbf{N}\| = \sqrt{3}$

- $f = 3z$  and  $z = 2 - x - y \Rightarrow f = 6 - 3x - 3y = 6 - 3r \cos \theta - 3 \sin \theta$ .
- $D : x^2 + y^2 \leq 1 = [0, 1] \times [0, 2\pi]$ .

$$\iint_S 3z dS = \int_0^{2\pi} \int_0^1 [6 - 3r \cos \theta - 3 \sin \theta] r dr d\theta = 6\pi.$$

**Example-2:**

Evaluate  $\iint_S (x^2 + y^2 + z^2) dS$ ,  $S$  is the part of the  $x + y + z = 2$  cylinder  $x^2 + y^2 = 9$  between the planes  $z = 0$  and  $z = 2$ , together with its top and bottom disks.

**Solution:** Let  $f = x^2 + y^2 + z^2$  and  $S = S_1 \cup S_2 \cup S_3$ . Then

$$\iint_S f dS = \iint_{S_1} f dS + \iint_{S_2} f dS + \iint_{S_3} f dS$$

i)  $S_1 : x^2 + y^2 = 9, \quad 0 \leq z \leq 2.$

$$\mathbf{N} = \langle 3 \cos u, 3 \sin u, 0 \rangle, \quad 0 \leq u \leq 2\pi, \quad 0 \leq v \leq 2$$

$$\|\mathbf{N}\| = 3 du dv \quad \text{and} \quad f = 9 + v^2$$

$$\iint_{S_1} f dS = \int_0^{2\pi} \int_0^2 (27 + 3v^2) dv du = 124\pi.$$

ii)  $S_2 : z = 0, \quad x^2 + y^2 \leq 9 \Rightarrow \|\mathbf{N}\| = \mathbf{j}, \quad \|\mathbf{N}\| = 1, \quad f = x^2 + y^2$

$$\iint_{S_2} f dS = \int_0^{2\pi} \int_0^3 (r^2) r dr d\theta = \frac{81\pi}{2}.$$

iii)  $S_3 : z = 2, \quad x^2 + y^2 \leq 9 \Rightarrow \|\mathbf{N}\| = \mathbf{j}, \quad \|\mathbf{N}\| = 1, \quad f = x^2 + y^2 + 4$

$$\iint_{S_3} f dS = \int_0^{2\pi} \int_0^3 (r^2 + 4) r dr d\theta = \frac{153\pi}{2}$$

$$\iint_S f dS = \iint_{S_1} f dS + \iint_{S_2} f dS + \iint_{S_3} f dS = 241\pi$$

**Example-3:**

Find the mass of a thin funnel in the shape of a cone  $z = \sqrt{x^2 + y^2}$ ,  $1 \leq z \leq 4$  if its density function is  $\sigma(x, y, z) = 10 - z$ .

**Solution:**  $\mathbf{N} = \left\langle \frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right\rangle$  and  $\|\mathbf{N}\| = \sqrt{2}$ .

- $\rho = 10 - z = 10 - r$  on  $S$ ,  $1 \leq r \leq 4, \quad 0 \leq \theta \leq 2\pi.$

- $m = \iint_S (10 - z) dS = \iint_D \sqrt{2} (10 - r) r dr d\theta$   
 $= \sqrt{2} \int_0^{2\pi} \int_1^4 (10r - r^2) dr d\theta = 118\sqrt{2}\pi$

- Let  $(\bar{x}, \bar{y}, \bar{z})$  be the center, where

$$\bar{x} = \frac{1}{m} \iint_S x \sigma dS = \frac{1}{118\pi} \int_0^{2\pi} \int_1^4 (10r - r^3) \cos \theta dr d\theta = 0$$

$$\bar{y} = \frac{1}{m} \iint_S y \sigma dS = \frac{1}{118\pi} \int_0^{2\pi} \int_1^4 (10r - r^3) \sin \theta dr d\theta = 0$$

$$\bar{z} = \frac{1}{m} \iint_S z \sigma dS = \frac{1}{118\pi} \int_0^{2\pi} \int_1^4 (10r - r^3) dr d\theta = \frac{17}{59}$$

17. **Surface Integral of a Vector-Field:**  $\iint_S \mathbf{F} \cdot \mathbf{n} dS$

- Let  $\mathbf{F} = \langle M, N, P \rangle$  be a continuous vector field defined on an oriented smooth surface  
 $S : \mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$ ,  $D : a \leq u \leq b, c \leq v \leq d$   
with unit normal vector  $\mathbf{n}$ . Then the **surface integral** of  $\mathbf{F}$  on  $S$ ,  
denoted by  $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ , is defined by

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} du dv \\ &= \int_a^b \int_c^d \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dv du \end{aligned}$$

- For  $S : z = g(x, y)$ ,  $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle$  is the upward (outward) normal vector and

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot \langle -z_x, -z_y, 1 \rangle dA,$$

$D$  = projection of  $S$  on the  $xy$ -plane.

- This integral is also called the **flux** of  $\mathbf{F}$  across  $S$ .

**Example-1:**

Find the flux of  $\mathbf{F} = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$  across the part of the plane  $x + 2y + z = 4$  in the first octant.

**Solution:**  $S : z = 4 - x - 2y$ ,  $D : 0 \leq y \leq 2, 0 \leq x \leq 4 - 2y$ .

- $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle = \langle 1, 2, 1 \rangle$  and  $\mathbf{F} \cdot \mathbf{N} = x + 2y - z = 2x + 4y - 4$

$$\begin{aligned} flux &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \int_0^2 \int_0^{4-2y} (2x + 4y - 4) dx dy \\ &= \int_0^2 (x^2 + 4xy - 4x) \Big|_0^{4-2y} dy = \int_0^2 (8y - 4y^2) dy = \frac{16}{3}. \end{aligned}$$

**Example-2:**

Find the flux of  $\mathbf{F} = xz\mathbf{i} - y\mathbf{k}$  across the part of the sphere  $x^2 + y^2 + z^2 = 4$  above the plane  $z = 1$ .

**Solution:**  $\phi = \cos^{-1} \left( \frac{z}{\rho} \right)$ , for  $z = 2$ ,  $\phi_1 = 0$ , for  $z = 1$ ,  $\phi_2 = \cos^{-1} \left( \frac{1}{2} \right) = \frac{\pi}{3}$

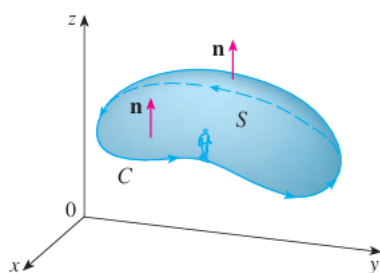
- $S : \mathbf{r}(u, v) = \langle 2 \sin u \cos v, 2 \sin u \sin v, 2 \cos u \rangle$ ,  $0 \leq v \leq 2\pi$ ,  $0 \leq u \leq \frac{\pi}{3}$
- $\mathbf{N} = 2 \sin u \mathbf{r}(u, v) = \langle 4 \sin^2 u \cos v, 4 \sin^2 u \sin v, 4 \sin u \cos u \rangle$
- $\mathbf{F}(\mathbf{r}(u, v)) = \langle 4 \sin u \cos u \cos v, 0, -2 \sin u \sin v \rangle$
- $\mathbf{F} \cdot \mathbf{N} = 16 \sin^3 u \cos u \cos^2 v - 8 \sin^3 u \cos u \sin v$   
 $= 8 \sin^3 u \cos u (1 + \cos 2v - \sin v)$

$$flux = \iint_S \mathbf{F} \cdot \mathbf{n} dS = 16\pi \int_0^{\frac{\pi}{3}} \sin^3 u \cos u du = 9\pi$$

18. **Stokes' Theorem:**  $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$

Let  $S$  be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve  $C$  with positive orientation. Let  $\mathbf{F}$  be a vector field whose components have continuous partial derivatives on an open region in  $\mathbb{R}^3$  that contains  $S$ . Then

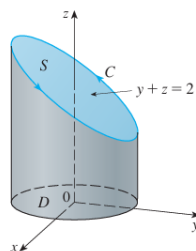
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$$



**Example-1:**

Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$ , where  $\mathbf{F} = \langle -y^2, x, z^2 \rangle$  and  $C$  is the curve of intersection of the plane  $y + z = 2$  and the cylinder  $x^2 + y^2 = 1$ . (Orient  $C$  to be counterclockwise when viewed from above.)

**Solution:**



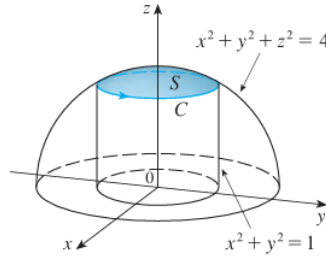
- The curve  $C$  is an ellipse.
- To evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$ , it's easier to use Stokes' Theorem. We first compute

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y^2 & x & z^2 \end{vmatrix} = \langle 0, 0, 1 + 2y \rangle$$

- $S$ : is the part of the plane  $z = 2 - x$  bounded by  $C$ .
- $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle = \langle 0, 1, 1 \rangle \Rightarrow \text{curl } \mathbf{F} \cdot \mathbf{N} = 1 + 2y = 1 - 2r \sin \theta$
- $D = \{(x, y) | x^2 + y^2 \leq 1\} = [0, 1] \times [0, 2\pi]$ -projection of  $S$  on the  $xy$ -plane.
- By Stokes' theorem,  $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \iint_D (1 + 2y) dA = \pi$ .

**Example-2:**

Use Stokes' Theorem to compute the integral  $\int_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$ , where  $\mathbf{F} = \langle xz, yz, xy \rangle$  and  $S$  is the part of the sphere  $x^2 + y^2 + z^2 = 4$  that lies inside the cylinder  $x^2 + y^2 = 1$  and above the  $xy$ -plane.

**Solution:**

- To find the boundary curve  $C$  we solve the equations  $x^2 + y^2 + z^2 = 4$  and  $x^2 + y^2 = 1$ .
- Subtracting, we get  $z = \sqrt{3}$  (since  $z > 0$ ).
- A vector equation of  $C$  is  $\mathbf{r}(t) = \langle \cos t, \sin t, \sqrt{3} \rangle$ ,  $0 \leq t \leq 2\pi$ .
- $\mathbf{r}'(t) = \langle -\sin t, \cos t, 0 \rangle$
- $\mathbf{F}(\mathbf{r}(t)) = \langle \sqrt{3} \cos t, \sqrt{3} \sin t, \sin t \cos t \rangle \Rightarrow \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0$
- By Stokes' theorem,  $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = 0$

**Example-3:**

Use Stokes' Theorem to compute the integral  $\int_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$ , where  $\mathbf{F} = \langle y^2 z^2, xz^2, xyz \rangle$  and  $S$  is the part of the paraboloid  $z = x^2 + y^2$  that lies inside the cylinder  $x^2 + y^2 = 4$ , oriented upward.

**Solution:**

- The boundary curve  $C$  is the intersection of  $z = x^2 + y^2$  and  $x^2 + y^2 = 4$ .
- Substituting, we get  $z = 4$ . That is, they intersect in the circle  $x^2 + y^2 = 4$  on the plane  $z = 4$ .
- A vector equation of  $C$  is  $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 4 \rangle$ ,  $0 \leq t \leq 2\pi$ .
- $\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t, 0 \rangle$
- $\mathbf{F}(\mathbf{r}(t)) = \langle 64 \sin^2 t, 32 \cos t, 16 \sin t \cos t \rangle$   
 $\Rightarrow \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -128 \sin^3 t + 64 \cos^2 t = 128(\cos^2 t - 1) \sin t + 32 + 32 \cos 2t$
- By Stokes' theorem,  

$$\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

$$= \int_0^{2\pi} [128 \cos^2 t \sin t - 128 \sin t + 32 + 32 \cos 2t] dt = 64\pi$$

19. **Divergence Theorem:**  $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_E \operatorname{div} \mathbf{F} dV$

Let  $E$  be a **simple solid region** and let  $S$  be the boundary surface of  $E$ , given with **positive (outward) orientation**  $\mathbf{n}$ . Let  $\mathbf{F}$  be a vector field whose component functions have continuous partial derivatives on an open region that contains  $E$ . Then

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_E \operatorname{div} \mathbf{F} dV \\ &= \iiint_E (M_x + N_y + P_z) dV \end{aligned}$$

**Example-1:**

Find the flux of the vector field  $\mathbf{F} = z\mathbf{i} + y\mathbf{j} + x\mathbf{k}$  over the unit sphere  $x^2 + y^2 + z^2 = 1$

**Solution:**  $\operatorname{div} \mathbf{F} = 1$

- $S$  is the bounder of the ball  $B : x^2 + y^2 + z^2 \leq 1$
- $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_B \operatorname{div} \mathbf{F} dV = \frac{4\pi}{3}$

**Example-2:**

Evaluate  $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ ,  $\mathbf{F} = xz\mathbf{i} + \sin(xz)\mathbf{j} + (z^2 + e^{xy^2})\mathbf{k}$  and  $S$  is the surface of the region  $E$  bounded by the parabolic cylinder  $y = 1 - x^2$  and the planes  $z = 0$ ,  $y = 0$  and  $y + z = 2$ .

**Solution:**  $\operatorname{div} \mathbf{F} = z + 0 + 2z = 3z$

- $S$  is the bounder of  $E = \{(x, y, z) | -1 \leq x \leq 1, 0 \leq y \leq 1 - x^2, 0 \leq z \leq 2 - y\}$

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_E \operatorname{div} \mathbf{F} dV = \int_{-1}^1 \int_0^{1-x^2} \int_0^{2-y} 3z dz dy dx \\ &= \frac{3}{2} \int_{-1}^1 \int_0^{1-x^2} (2-y)^2 dy dx = -\frac{1}{3} \int_{-1}^1 [(1+x^2)^3 - 8] dx \\ &= -\int_0^1 [x^6 + 3x^4 + 3x^2 - 7] dx = \frac{184}{35} \end{aligned}$$